# Yonghoon Jeong

Republic of Korea Naval Academy

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## Research Interests

Wireless Communication Systems
Performance Analysis and Modeling
Physical-Layer Wireless Sensing and Networking
Low-Power Wide-Area Networks (LPWANs)
Backscatter Communication and Networking
Ocean Internet of Things (Ocean IoT)

## Education

**M.S. in Electrical and Computer Engineering** 2024 - 2026
- Seoul National University, Seoul, Republic of Korea
- Advisor: Prof. Saewoong Bahk (currently President of ETRI)

**B.S. in Electronics and Communications Engineering** 2018 - 2023
- Kwangwoon University, Seoul, Republic of Korea

## Professional Experience

**Republic of Korea Naval Academy (ROKNA)** 2026 - Present
- Full-time Lecturer, Department of Cyber Science
- Researcher, Institute of Naval Research
- Commissioned as a Republic of Korea Navy officer; mandatory military service ends in May 2029.

**Seoul National University (SNU)** 2024 - 2026
- Researcher, Network Laboratory (NETLAB), Institute of New Media and Communications (INMC)

**Electronics and Telecommunications Research Institute (ETRI)** 2023
- Researcher, AI-SoC Infrastructure Research Section

**Korea Advanced Institute of Science and Technology (KAIST)** 2022
- Research Intern, Unmanned Systems Research Group, School of Electrical Engineering (EE)
- Advisor: Prof. David Hyunchul Shim

## Publications

### Technical Articles

- **Yonghoon Jeong**, "A Survey of Underwater Backscatter Networks and Their Naval Applications," **to appear** in ***Naval Ship Technology Journal***, vol. 95, Dec. 2026. **(Best Paper Award)**

### International Papers

- "On the Symbol Error Rate of Underwater Acoustic Backscatter Systems," **under review** at ***IEEE Transactions on Communications***.
- "Underwater Acoustic Backscatter: Planning for Complete Data Collection," **under review** at ***IEEE Communications Magazine***.
- "Characterizing the Relation Between the Energy Scattering Effect and Symbol Error Rate in Non-Linear Chirp Modulation for LoRa," **under review** at ***IEEE Wireless Communications Letters***.
- "Symbol Error Rate Analysis and Approximation for Arbitrary Chirp Modulation Under Multipath, Doppler, and Interference," **under review** at ***IEEE Transactions on Communications***.
- "On the Symbol Error Rate of Non-Linear Chirp Modulation for LoRa Under Asynchronous Interference," **under review** at ***IEEE Wireless Communications Letters***.
- **Yonghoon Jeong**, Youngwook Son, and Saewoong Bahk, "On the Interference Resilience of Non-Linear Chirp Modulation for LoRa: An Analytical Insight into Energy Scattering Effect," **published** in ***IEEE Communications Letters***, vol. 30, pp. 2238-2242, Jun. 2026. **(Paper)**

- Dongrack Choi, Yubin Choi, **Yonghoon Jeong**, Jeongyeup Paek, and Saewoong Bahk, "D²-PLC: Holistic Design and Implementation of High-Data-Rate Duplex DC Power Line Communication Network," **published** in ***IEEE Internet of Things Journal***, vol. 12, pp. 45325-45340, Nov. 2025. **(Paper)**
- Dongrak Choi, **Yonghoon Jeong**, Yubin Choi, and Saewoong Bahk, "Demo: A Programmable High-Throughput Duplex DC-PLC Testbed for Power and Data Integration," **published** in the proceedings of ***IEEE ICNP 2025***, Seoul, Republic of Korea, Sep. 22-25, 2025. **(Paper)**
- **Yonghoon Jeong** and Youngseok Choi, "Diffusion-Phys: Noise-Robust Heart Rate Estimation from Facial Videos via Diffusion Models," **published** in ***Biomedical Engineering Letters***, vol. 15, pp. 575-585, Apr. 2025. **(Paper)**

### Domestic Papers

- **Yonghoon Jeong** and Saewoong Bahk, "A Study on Direct Current Power Line Communication (DC-PLC) with Applications in Automotive, Maritime, and Aviation," **published** in the proceedings of ***The Conference of the Korea Institute of Communications and Information Sciences (KICS)***, Gyeongju, Republic of Korea, Nov. 20-22, 2024.
- **Yonghoon Jeong**, "Development of Non-contact Heart Rate Measurement Technology Based on Facial Video," **published** in the proceedings of ***The Conference of the Korean Institute of Information Scientists and Engineers (KIISE)***, Jeju, Republic of Korea, Jun. 23-25, 2021.

## Academic Service

**Journal Reviewer**

- *IEEE Internet of Things Journal*

## Awards & Scholarships

**Best Paper Award (Rear Admiral's Commendation)** 2026
- *Naval Ship Technology Journal*, Naval Technology Research Institute, Republic of Korea Navy

**BK21 Scholarship** 2024-2025
- Brain Korea 21 (BK21), National Research Foundation of Korea

**KOSFFL Scholarship** 2022-2023
- Korea Scholarship Foundation for the Future Leaders (KOSFFL)

**Excellent Paper Award** 2021
- Korea Computer Conference 2021

**University President's Award** 2020
- Seoul Capstone Design, Seoul

**Samsong Scholarship** 2020
- Samsong Scholarship Foundation

**Dean's List** 2020
- Highest GPA

## Teaching Experience

**Creative Engineering Design** 2024
- Teaching Assistant; Prof. Saewoong Bahk

**Electronic Circuits Experiment I** 2023
- Instructor; Emeritus Prof. Young-chul Chung

**Electronic Circuits Experiment II** 2023
- Instructor; Emeritus Prof. Young-chul Chung

**Electronic Circuits Experiment I** 2022
- Instructor; Prof. Youngseok Choi

## Technical Skills

**Programming Languages:** C, C++, Python, MATLAB

**Frameworks & Libraries:** ROS, ROS2, TensorFlow, PyTorch

**Simulation Tools:** MATLAB/Simulink, ns-3

**Development Tools:** Linux, Git, Docker